

A SQUARE-ZERO UNITIZATION WITH NON-OPEN MULTIPLICATION

AGENT LANE 05

SOURCE AND TARGET

The source paper is Draga–Kania, *When is multiplication in a Banach algebra open?*, arXiv:1704.08608 [1]. Open Problem 3 on page 14 asks:

Do commutative, unital Banach algebras with zero-dimensional maximal ideal space have (uniformly) open multiplication?

What if idempotents in the algebra are linearly dense?

- (3) Do commutative, unital Banach algebras with zero-dimensional maximal ideal space have (uniformly) open multiplication? What if idempotents in the algebra are linearly dense?

RESULT

The first question has a negative answer as written. There is a commutative unital complex Banach algebra whose maximal ideal space is a singleton, but whose multiplication is not open. Consequently its multiplication is not uniformly open.

The example is non-semisimple and has only the trivial idempotents 0 and 1. Thus it does not answer the stronger follow-up question about linearly dense idempotents.

PROOF INTUITION

Take the unitization of a square-zero two-dimensional ideal. Since every element of the ideal is nilpotent, every character vanishes on that ideal, so the maximal ideal space is just one point. Openness fails for a geometric reason. Near a fixed nilpotent vector e_1 , both factors have ideal parts close to e_1 . To produce a product with scalar coordinate 0, one of the scalar coordinates must be exactly 0. The ideal part of the product then becomes a scalar multiple of a vector still close to e_1 , and so it cannot equal a small nonzero multiple of a transverse vector e_2 .

Theorem 1. *There exists a commutative unital complex Banach algebra A such that $\text{Max}(A)$ is zero-dimensional and multiplication*

$$A \times A \longrightarrow A, \quad (a, b) \mapsto ab,$$

is not open.

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Proof. Let $E = \mathbb{C}^2$, equipped with the norm $\|(z_1, z_2)\|_\infty = \max\{|z_1|, |z_2|\}$. Let

$$A = \mathbb{C} \oplus E$$

with norm $\|(\lambda, x)\| = |\lambda| + \|x\|_\infty$ and multiplication

$$(\lambda, x)(\mu, y) = (\lambda\mu, \lambda y + \mu x).$$

This is the usual unitization of the square-zero algebra E , where $xy = 0$ for all $x, y \in E$. The displayed norm is submultiplicative:

$$\begin{aligned} \|(\lambda, x)(\mu, y)\| &= |\lambda\mu| + \|\lambda y + \mu x\|_\infty \\ &\leq |\lambda||\mu| + |\lambda| \|y\|_\infty + |\mu| \|x\|_\infty \leq \|(\lambda, x)\| \|(\mu, y)\|. \end{aligned}$$

The algebra is plainly commutative and unital, with identity $(1, 0)$.

We first identify the maximal ideal space. If $\varphi : A \rightarrow \mathbb{C}$ is a character and $x \in E$, then $(0, x)^2 = 0$, hence

$$\varphi(0, x)^2 = \varphi((0, x)^2) = 0.$$

Thus $\varphi(0, x) = 0$ for every $x \in E$, and since $\varphi(1, 0) = 1$,

$$\varphi(\lambda, x) = \lambda.$$

So A has exactly one character. Its maximal ideal space is therefore a singleton, hence zero-dimensional.

It remains to prove that multiplication is not open. Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ in E , and put

$$a = (0, e_1) \in A.$$

We show that multiplication is not open at (a, a) . Fix $0 < \varepsilon < 1$. We claim that no ball around $0 = a^2$ is contained in

$$B_A(a, \varepsilon) \cdot B_A(a, \varepsilon).$$

Indeed, let $\delta > 0$ and choose $0 < t < \delta$. Consider

$$z = (0, te_2) \in A.$$

Then $\|z\| = t < \delta$. Suppose, towards a contradiction, that

$$z = (\alpha, x)(\beta, y)$$

with $(\alpha, x), (\beta, y) \in B_A(a, \varepsilon)$. Comparing scalar coordinates gives

$$\alpha\beta = 0.$$

Hence $\alpha = 0$ or $\beta = 0$. If $\alpha = 0$, then the ideal coordinate of the product is $\beta x = te_2$. Since $t \neq 0$, we have $\beta \neq 0$, and therefore $x \in \mathbb{C}e_2$. But every vector in $\mathbb{C}e_2$ has distance at least 1 from e_1 in the supremum norm:

$$\|x - e_1\|_\infty \geq 1.$$

This contradicts

$$\|(\alpha, x) - a\| = |\alpha| + \|x - e_1\|_\infty < \varepsilon < 1.$$

The case $\beta = 0$ is identical, using y instead of x .

Thus every ball around 0 contains points not belonging to $B_A(a, \varepsilon) \cdot B_A(a, \varepsilon)$. Multiplication is not open at (a, a) , and hence is not an open map. $\square$

VERIFICATION NOTES

No computational verification is involved. The key algebraic checks are:

- $0 \oplus E$ is a square-zero ideal, so all characters kill it;
- $\text{Max}(A)$ is the singleton scalar character;
- the failure of openness is witnessed by the missing transverse points $(0, te_2)$ in every neighborhood of 0.

NOVELTY AND SEARCH BOUNDS

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1704.08608, the title, “zero-dimensional maximal ideal space”, “square-zero”, “unitization”, and “open multiplication”. No existing packet for this target was found.

External web searches on 2026-06-15 used exact and near-exact phrases including “zero-dimensional maximal ideal space open multiplication Banach algebra”, “square-zero open multiplication Banach algebra”, “unitization zero-dimensional maximal ideal space multiplication open”, and the exact printed question from Open Problem 3. These searches did not find a later explicit answer. Novelty confidence is medium because the example is elementary and may be known folklore.

HUMAN REVIEW RECOMMENDATION

Review as a candidate counterexample to the first clause of Open Problem 3 as printed. The main semantic issue is whether the authors intended an unstated semisimplicity assumption. The dense-idempotent follow-up remains open in this packet.

REFERENCES

- [1] S. Draga and T. Kania, *When is multiplication in a Banach algebra open?*, arXiv:1704.08608, 2017.

Human Review: Square-zero unitization and non-open multiplication

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed example appears to correctly answer the first clause of Open Problem 3 in Dragă-Kania as printed. Namely, it gives a commutative unital Banach algebra whose maximal ideal space is zero-dimensional, but whose multiplication map is not open.

The construction is the unitization of a square-zero ideal. Let

$$A = \mathbb{C} \oplus E, \quad E = \mathbb{C}^2,$$

with multiplication

$$(\lambda, x)(\mu, y) = (\lambda\mu, \lambda y + \mu x)$$

and norm $\|(\lambda, x)\| = |\lambda| + \|x\|_\infty$. The algebraic and norm checks are straightforward: A is a commutative unital Banach algebra, and the norm is submultiplicative.

Maximal Ideal Space

The computation of the maximal ideal space is correct. If $\varphi : A \rightarrow \mathbb{C}$ is a character and $x \in E$, then

$$(0, x)^2 = 0.$$

Therefore

$$\varphi(0, x)^2 = \varphi((0, x)^2) = 0,$$

so $\varphi(0, x) = 0$. Since any character on a unital complex Banach algebra satisfies $\varphi(1) = 1$, it follows that

$$\varphi(\lambda, x) = \lambda.$$

Thus A has exactly one character, so $\text{Max}(A)$ is a singleton and hence zero-dimensional.

Failure of Openness

The failure of openness is also correct. Let $e_1 = (1, 0)$, $e_2 = (0, 1)$ in E , and put

$$a = (0, e_1).$$

Then $a^2 = 0$. To show that multiplication is not open at (a, a) , it is enough to find $0 < \varepsilon < 1$ such that

$$B_A(a, \varepsilon) \cdot B_A(a, \varepsilon)$$

is not a neighbourhood of 0.

Given any $\delta > 0$, choose $0 < t < \delta$ and consider

$$z = (0, te_2).$$

Then $\|z\| < \delta$. Suppose, for contradiction, that

$$z = (\alpha, x)(\beta, y)$$

with both factors in $B_A(a, \varepsilon)$. Comparing scalar coordinates gives

$$\alpha\beta = 0.$$

Hence $\alpha = 0$ or $\beta = 0$. If $\alpha = 0$, then the ideal coordinate equation becomes

$$\beta x = te_2.$$

Since $t \neq 0$, one has $\beta \neq 0$, and so $x \in \mathbb{C}e_2$. But every vector in $\mathbb{C}e_2$ is at distance at least 1 from e_1 in the supremum norm:

$$\|x - e_1\|_\infty \geq 1.$$

This contradicts the assumption that

$$(\alpha, x) \in B_A(a, \varepsilon), \quad \varepsilon < 1.$$

The case $\beta = 0$ is identical, using y instead of x .

Thus every ball around $0 = a^2$ contains points $(0, te_2)$ which are not in the product of the two ε -balls around a . Therefore multiplication is not open.

Review Outcome

The proof appears correct. It is a simple and effective counterexample to the first clause of the open problem as literally stated.

The main caveat is semantic rather than mathematical: the example is highly non-semisimple, and it has only the trivial idempotents. Thus it does not answer the stronger dense-idempotent follow-up, and it may be worth asking the authors whether they intended an additional assumption such as semisimplicity. As written, however, the counterexample is valid.